Coding & Solution

Date	13 July 2026
Team ID	
Project Name	HDI Predictor — Human Development Index Estimation & Recommendation Web App
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of the implemented HDI Predictor web application, built against the problem statement: given a country's life expectancy, education attainment, and income (GNI per capita), the system predicts a Human Development Index tier — Very High, High, Medium, or Low — and recommends where the country should improve.

Solution Summary

Repository Link / URL	[Add your GitHub / GitLab repository link here]
Programming Language(s)	HTML5, CSS3, JavaScript (ES6)
Framework(s) Used	None required — vanilla JS single-page app. (Optional backend extension: Flask or Node/Express for persistence and multi-country storage.)
Key Features Implemented	• Three interactive input controls: Life Expectancy, Education (Mean + Expected Years of Schooling), and GNI per capita (PPP) • Real-time computation of Life Expectancy Index, Education Index, and Income Index using the UNDP geometric-mean methodology • Automatic classification into four tiers: Very High (≥ 0.800), High (0.700–0.799), Medium (0.550–0.699), Low (< 0.550) • Live 'development triangle' visualization plotting the three sub-indices • Dynamic recommendation engine that identifies the weakest of the three pillars and generates tailored improvement guidance • Fully interactive, no page reloads / no server calls — instant feedback as sliders move • Responsive layout for desktop and mobile; keyboard-accessible controls
Pending / Incomplete Features	• No backend/database — country data is entered manually rather than fetched from a live dataset (e.g., World Bank / UNDP API) • Classification is formula-based (matches the official HDI methodology) rather than a trained ML model; a supervised model could be layered on top later • No user accounts, saved history, or PDF/report export yet • No multi-country comparison view
Setup / Run Instructions	• Download hdi_predictor.html • Open it directly in any modern browser (Chrome, Edge, Firefox, Safari) — no installation or server required

	• Move the sliders for life expectancy, schooling years, and GNI per capita to see the predicted tier and recommendation update live

Code Quality Checklist

S.No	Criteria	Status
1	Code is modular and organized into functions / classes	**Yes**
2	Meaningful variable and function names are used	**Yes**
3	Code includes comments / documentation where necessary	**Yes**
4	Error handling is implemented for critical operations (input clamping to valid ranges)	**Yes**
5	The application runs without critical errors	**Yes**
6	Code is committed to a version control repository	**Pending**

Reviewer Evaluation — Scenario Testing

As project reviewer, the model was exercised against the three scenarios defined in the problem statement, confirming it produces the expected tier and a coherent, actionable recommendation in each case.

Scenario	Life Exp.	Schooling	GNI per capita	Predicted HDI Tier	Weakest Pillar Flagged
1 — Very High Development	82 yrs	Mean 13 / Exp 17	$55,000	**Very High**	None — balanced
2 — Development Gaps (emerging economy)	68 yrs	Mean 7 / Exp 12	$9,500	**Medium**	Education
3 — Requires Intervention	54 yrs	Mean 3 / Exp 6	$1,800	**Low**	Education

Prediction & Recommendation Logic

The application follows the same three-step logic a human reviewer would apply:

- Normalize each raw input into a 0–1 index (Life Expectancy Index, Education Index, Income Index) using the fixed UNDP minimum/maximum goalposts.
- Combine the three indices with a geometric mean, so no single strong pillar can fully compensate for a weak one — the app reflects this by never rewarding lopsided profiles.
- Classify the composite score into Very High / High / Medium / Low, and separately identify the lowest-scoring pillar to generate a targeted, human-readable improvement recommendation (health investment, education access, or income growth).

Additional Notes / Comments

The interface covers all states described in the problem statement (Very High, High, Medium, Low) through a single continuous model rather than hard-coded cases, so any combination of inputs — not just the three sample scenarios — produces a consistent, explainable result. Next iteration: connect to a live World Bank / UNDP data source and persist a repository link once version control is set up.